

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5122

5 Credits: 4

Program Core

Source-grounded

Wood Products-II

□ To provide basic knowledge regarding the manufacturing process of Wood based

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5122 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5122.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5122
Course title	Wood Products-II
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide basic knowledge regarding the manufacturing process of Wood based
- panel products such as Block board, particle board, fibre board, oriented strand
- To make aware of the details of their properties and testing.

Course outcomes

CO	Official outcome	Study level
CO1	15 Applying testing and properties. Disseminate pressing, finishing, testing and	See official syllabus
CO2	14 Applying properties of particle board Explain the Chipping of wood, pulping methods,	See official syllabus
CO3	Mat formation manufacture of Insulation board and 15 Applying Hardboard Explain sandwich composition, bulking treatments	See official syllabus
CO4	14 Applying and testing and properties of wood laminates	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 1 2
CO2	CO2 3 2
CO3	CO3 3
CO4	CO4 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	properties	4	As specified
Module 2	Disseminate pressing, finishing, testing and properties.	4	As specified
Module 3	manufacture of Insulation board and Hardboard	4	As specified
Module 4	properties of wood laminates.	3	As specified

MODULE 1

properties

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: properties
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Remembering board Explain exterior grade, interior grade, single

3 Remembering board Explain exterior grade, interior grade, single is an official topic in Module 1 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering board Explain exterior grade, interior grade, single using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering board explain exterior grade, interior grade, single should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Remembering board Explain exterior grade, interior grade, single means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 3 Remembering board Explain exterior grade, interior grade, single definition/meaning. steps, application. Technical terms English- .

▼ layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B

layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B is an official topic in Module 1 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, layer, three layer, multi-layer, graded, 7 remembering extrusion pressed and flat presses p.b explain the schematic outline for p.b should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

▼ 3 Understanding manufacture

3 Understanding manufacture is an official topic in Module 1 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding manufacture using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding manufacture should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding manufacture means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding manufacture

പ്രദേശം പ്രദേശം definition/meaning പ്രദേശം പ്രദേശം. പ്രദേശം പ്രദേശം പ്രദേശം, steps, application പ്രദേശം പ്രദേശം പ്രദേശം. Technical terms English-പ്രദേശം പ്രദേശം പ്രദേശം.

▼ Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.

Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. is an official topic in Module 1 of Wood Products-II. Study the terminology first, then

connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline need of drying of pb 2 applying block board types - block method, strip method. assembling of block board core. continuous gluing, spot gluing. decorative and non-decorative block board flush door types - solid core flush door, cellular core (cellulose) flush door. adhesive used. chipping and rebate cutting, facing testing and properties of block board & flush door properties of block board & flush door. alternative wetting and drying test, knife test, boiling test of blackboard-particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. classification of p.b - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed p.b uses of p.b, raw materials , adhesive - uf and pf resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, schematic out line p.b name factors. particle preparation, debarking cross cutting, splitting classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard- Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer , multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.

□□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Disseminate pressing, finishing, testing and properties.

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Disseminate pressing, finishing, testing and properties.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip

Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline chip storage and drying of chip 2 understanding summaries general considerations of chip should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഔ Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip ഓരോന്നും definition/meaning ഓരോന്നും. ഓരോന്നും ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നും ഓരോന്നും. Technical terms English-ഔ ഓരോന്നും ഓരോന്നും.

▼ 5 Applying blending

5 Applying blending is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying blending using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying blending should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying blending means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-5 Applying blending
 definition/meaning . steps, application . Technical terms English-
 .

▼ Explain particle board 2 Understanding Enumerate the properties and testing of

Explain particle board 2 Understanding Enumerate the properties and testing of is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain particle board 2 Understanding Enumerate the properties and testing of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain particle board 2 understanding enumerate the properties and testing of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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▼ **5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation,**

5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric

and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation, is an official topic in Module 2 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 remembering particle board storage of chips - horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. chip blending - definition, general consideration, batch operation, continuous operation, mat laying - terms and failures in volumetric and gravimetric dosing. dosing of resinated chips. volumetric and gravimetric dosing. dosing of resinated chips volumetric dosing and control of the weight of the chips. spreader heads and spreading machines, resultant mat- pre-pressing and pressing of p.b - pre pressing methods - advantages and its disadvantages. pressing - two wetting methods of mat, four particular proceedings of the hot platen press. three methods to achieve short pressing time in single day light press- effect of press platen temperature on physical properties of p.b. effect of curing time on physical properties of p.b, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of p.b. characteristics of uf resin with pf resin. list 3 caul less pressing process - continuous belt system. chip craft vertical extrusion process, lane wood horizontal extrusion processes calendar particleboard - mande - baisen system - p.b finishing - sizing sanding or thickening by wide belt sanding machine. physical and chemical changes during storage of p.b after leaving press. properties and testing of p.b - preparation and conditioning of test specimen, factors affecting the properties of p.b. physical properties, general properties, mechanical properties, technological properties and resistance against destruction. p.b testing methods - density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, modulus of elasticity, tensile strength perpendicular to the plane of board. state standard test result of p.b. explain the chipping of wood, pulping methods, mat formation, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading

Aspect	What to prepare
	machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general

consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation, $\frac{1}{2}$ definition/meaning $\frac{1}{2}$. $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$, steps, application $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$. Technical terms English- $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

manufacture of Insulation board and Hardboard

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: manufacture of Insulation board and Hardboard
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 6 Understanding manufacturing process of fiberboard

6 Understanding manufacturing process of fiberboard is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding manufacturing process of fiberboard using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding manufacturing process of fiberboard should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding manufacturing process of fiberboard means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-6 Understanding manufacturing process of fiberboard
 definition/meaning
 steps, application
 Technical terms English-

▼ Explain pulping processes

Explain pulping processes is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain pulping processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain pulping processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain pulping processes means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നു പറയുന്ന പൾപ്പിംഗ് പ്രക്രിയകൾ പരിശോധിക്കുക. പൾപ്പിംഗ് definition/meaning എന്താണ്. പൾപ്പിംഗ് പ്രക്രിയയുടെ steps, application എന്താണ്. Technical terms English-ൽ എഴുതുക.

▼ Describe mat formation

Describe mat formation is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe mat formation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe mat formation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe mat formation means in the official course context
Study lens	Principle → components or stages → result → application

▼ Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and

Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification

- testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and is an official topic in Module 3 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop testing of hardboard and mdf 3 applying fiber board insulation board mdf and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the m.c of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - fibreboard - block diagram of hardboard manufactures - sizing - mat formation and pressing - oil tempering, heat treatment, humidification - testing methods of fiber board. explain sandwich composition, bulking treatments and testing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing

Aspect	What to prepare
	methods of fiber board. Explain sandwich composition, bulking treatments and testing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□ □□□□□ □□□□□□□, steps,

application ഏകദേശം അനുസരിച്ചുള്ളതാണ്. Technical terms English-ൽ അനുസരിച്ചുള്ളതാണ്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

properties of wood laminates.

This module is presented from the official syllabus scope for Wood Products-II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: properties of wood laminates.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Oriented strand board

Oriented strand board is an official topic in Module 4 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Oriented strand board using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, oriented strand board should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Oriented strand board means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓറിയന്റഡ് സ്ട്രാൻഡ് ബോർഡ് (OSB) ന്റെ definition/meaning, ഉപയോഗം, steps, application എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ പരിഭാഷിക്കുക.

▼ Outline Densified wood

Outline Densified wood is an official topic in Module 4 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Densified wood using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline densified wood should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Densified wood means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓൗട്ലൈൻ ഡെൻസിഫൈഡ് വുഡ് ഏകദേശം
 ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps,
 application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഓ ഏകദേശം
 ഏകദേശം.

▼ Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking - Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link
https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1
http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 2
http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 3
https://www.fpl.fs.fed.us/documents/fplgtr/fpl_gtr190.pdf 4
<http://www.fao.org/3/an647e/an647e00.pdf> 5

Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity,

heat stabilisation & cross linking - Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link

https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1

http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3

http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4

https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5

<http://www.fao.org/3/an647e/an647e00.pdf> is an official topic in Module 4 of Wood Products-II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg -

Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link
https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1 cts
http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3
http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4
https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5
<http://www.fao.org/3/an647e/an647e00.pdf> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sandwich composition 5 applying oriented strand board - manufacturing methods - wood wool board - method of manufacture of wood wool board - properties and uses - cement bonded particleboard - method of manufacture - moulded particle board - werzalit process - densified wood - method of preparation of raw materials - densification of wood by impregnation and compression - metalized wood - compreg - dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-define sandwich composition - explain the raw materials - edge banding of particle board -

laminated pb - bulking treatment - bulking treatments - bulking treatments such as salt treatment, sugar treatment, poly ethylene glycol treatment, wax treatment, phenolic resin treatment - properties and uses of impreg - acetylated wood - testing and properties. text / reference: t/r book title/author franz. f.p kollmann, principles of wood science and technology vol. ii - t1 springer thomas m malon, modern particle board and drying process of fiber board t2 manufacturing online resources: sl. no website link

https://www.researchgate.net/publication/265168265_timber_and_wood_produ 1

cts http://www.kstr.lth.se/fileadmin/kstr/pdf_files/timber_engineering_2017/timber_and_engineered_wood_products.pdf 3

http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4

https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5

<http://www.fao.org/3/an647e/an647e00.pdf> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sand witch Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: 5 No Website Link https://www.researchgate.net/publication/265168265_Timber_and_wood_proc 1 cts http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3 http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4

Aspect	What to prepare
	https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5 http://www.fao.org/3/an647e/an647e00.pdf means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain sandwich composition 5
 Applying Oriented Strand board - Manufacturing Methods - wood wool board
 - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board -
 Werzalit Process - Densified Wood - method of Preparation of raw materials -
 densification of Wood by Impregnation and Compression - Metalized Wood -
 Compreg - Dimensional stabilisation of wood - the dimensional stabilization
 such as cross lamination, external and Internal coating, reduction in
 hygroscopicity, heat stabilisation & cross linking-Define sand witch
 Composition - Explain the raw materials - Edge banding of Particle Board -
 Laminated PB - Bulking treatment - bulking treatments - Bulking treatments
 such as salt treatment, Sugar treatment, poly ethylene glycol treatment,
 wax treatment, Phenolic resin treatment - Properties and uses of impreg -
 acetylated wood - Testing and Properties. Text / Reference: T/R Book
 Title/Author Franz. F.P Kollmann, Principles of wood science and technology
 Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying
 process of fiber board T2 manufacturing Online resources: SI. No Website
 Link
https://www.researchgate.net/publication/265168265_Timber_and_wood_produ

1 cts

http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3

http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4

https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5

<http://www.fao.org/3/an647e/an647e00.pdf> മരംകയറ്റം, മരംകയറ്റം

definition/meaning മരംകയറ്റം. മരംകയറ്റം മരംകയറ്റം, steps,

application മരംകയറ്റം മരംകയറ്റം മരംകയറ്റം. Technical terms English-മരംകയറ്റം

മരംകയറ്റം.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL**Module-wise questions and answers**

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 3 Remembering board Explain exterior grade, interior grade, single. (3 marks)**

► **Define or explain layer, three layer, multi-layer, graded, 7 Remembering extrusion pressed and flat presses P.B Explain the schematic outline for P.B. (3 marks)**

► **Define or explain 3 Understanding manufacture. (3 marks)**

► **Define or explain Outline Need of Drying of PB 2 Applying Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer, multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials, adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic outline P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle**

dryers, burned waste gas stream dryer, suspension type dryer.. (3 marks)

Module 2

► Define or explain Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip. (3 marks)

► Define or explain 5 Applying blending. (3 marks)

► Define or explain Explain particle board 2 Understanding Enumerate the properties and testing of. (3 marks)

► Define or explain 5 Remembering particle board Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the

properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B. Explain the Chipping of wood, pulping methods, mat formation,. (3 marks)

Module 3

► Define or explain 6 Understanding manufacturing process of fiberboard. (3 marks)

► Define or explain Explain pulping processes. (3 marks)

► Define or explain Describe mat formation. (3 marks)

► Define or explain Develop testing of hardboard and MDF 3 Applying Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment,

Humidification - testing methods of fiber board. Explain sandwich composition, bulking treatments and testing and. (3 marks)

Module 4

► **Define or explain Oriented strand board. (3 marks)**

► **Define or explain Outline Densified wood. (3 marks)**

► **Define or explain Explain sandwich composition 5 Applying Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking- Define sand witch Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties. Text / Reference: T/R Book Title/Author Franz. F.P Kollmann, Principles of wood science and technology Vol. II - T1 Springer Thomas M Malon, Modern particle board and drying process of fiber board T2 manufacturing Online resources: Sl. No Website Link
https://www.researchgate.net/publication/265168265_Timber_and_wood_products 1 cts
http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf 3
http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf 4
https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf 5
<http://www.fao.org/3/an647e/an647e00.pdf>. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Remembering board Explain exterior grade, interior grade, single.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Outline Chip storage and Drying of Chip 2 Understanding Summaries general considerations of chip.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **6 Understanding manufacturing process of fiberboard.**

Module 4

1-mark definition: State the meaning of **Oriented strand board.**

3-mark explanation: Explain one principle, process or comparison from this module.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link
- https://www.researchgate.net/publication/265168265_Timber_and_wood_produ
- http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf
- http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf
- https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf
- <http://www.fao.org/3/an647e/an647e00.pdf>

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